



Note: Maximum time: 2 Hours. Total Marks: 40

- Answer all of the following questions. Answers must be brief and to the point.
 - Make reasonable assumptions wherever necessary and state them.
1. (a) [3 points] Explain the different types of Cloud Storages with examples.
(b) [2 points] What are the roles of an user and provider for a Software as a Services (SaaS)?
 2. (a) [3 points] A company needs to handle massive video streaming traffic worldwide, scaling during peak times (e.g., new show releases). Managing its own data centers was costly and limited. Explain how public cloud based deployment help here.
(b) [2 points] What do you mean by Virtual Machine Snapshot?
 3. (a) [3 points] Explain the Memory Transfer Overheads that are present when a VM migration is performed.
(b) [2 points] What are the advantages of a Hypervisor? Explain.
 4. (a) [3 points] Write a python program using boto to add an inbound rule in a security group.
(b) [2 points] Why do we need to perform deployment refinement for a cloud application?
 5. (a) [3 points] How Role-Based Access Control (RBAC) improves the Identity and Access Control in cloud? Explain.
(b) [2 points] What do you mean by Southbound Interface and Northbound Interface in Software Defined Networking?
 6. (a) [3 points] How can map reduce help in big data processing? Give an example.
(b) [2 points] What are the different components of service level agreement (SLA)?
 7. (a) [2 points] Explain the functionality of three OpenStack components.
(b) [3 points] How Amazon Simple Queue Service (SQS) works? Can it reduce tight coupling?
 8. (a) [3 points] A company wants to develop a blog platform where users can: (i) Create, edit, and delete blog posts; (ii) View posts; (iii) Comment on posts; (iv) Like posts. How can you design this system using MVC Architecture?
(b) [2 points] What factors would you consider when you design a Cloud Application? Explain.